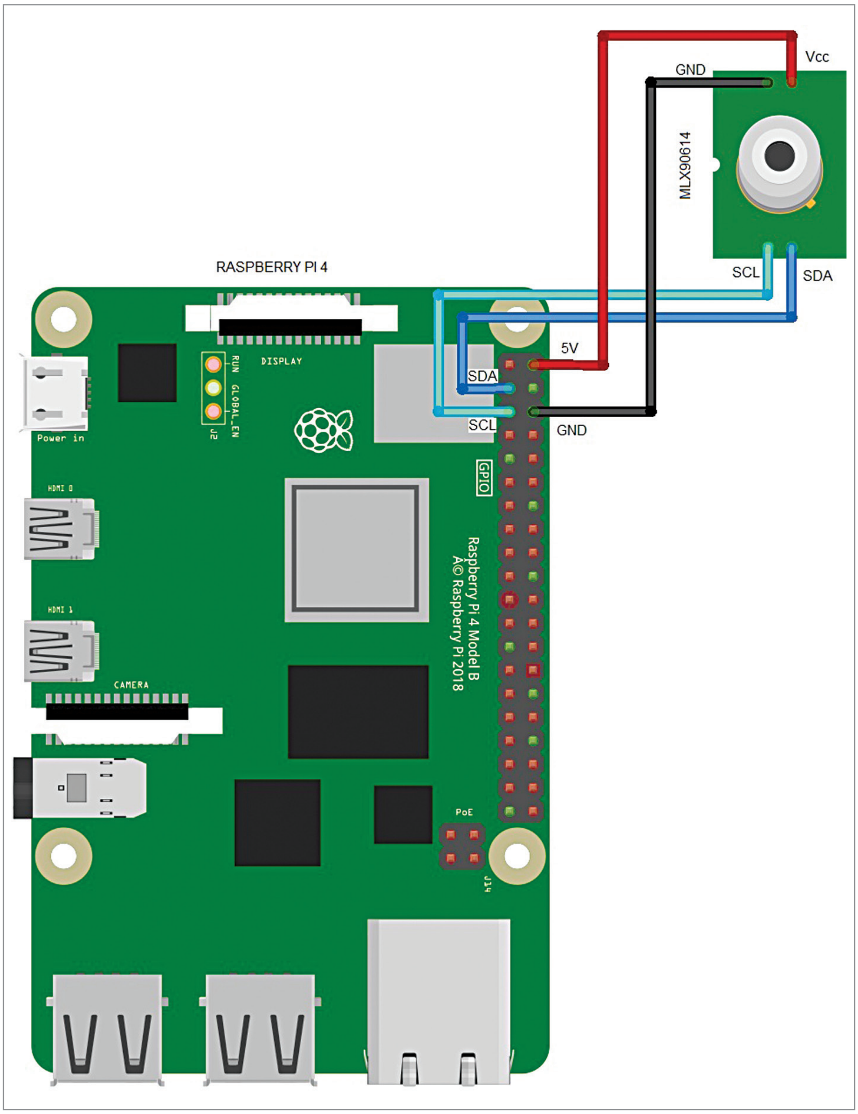

Fig. 8: Connection of MLX90614 with RPI

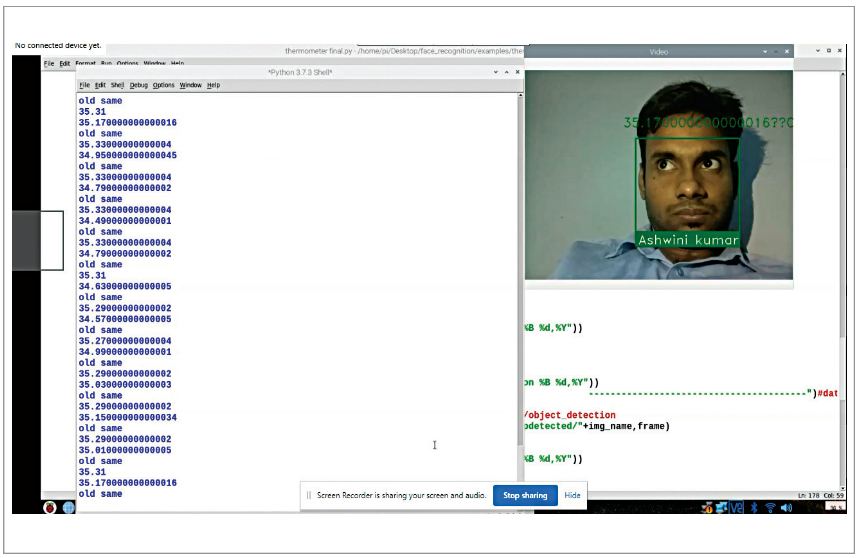

Fig. 9: Detecting face and temperature

an array having the list of names and images of all identified people. The names and images need to be in the same order. If we do not have an image of a person in the database, it is considered as unknown.

The video frame will be processed in the while loop, which will recognise the name of the person and then match it with the respective face that was detected. If it is different, it will use the MLX90614 sensor to measure the temperature of the person. Thereafter, using espeak, the name and temperature of that person will be announced.

The if() statements that we had written check whether the temperature is greater than 37 degrees Celsius or not. If the temperature detected is greater, the system tells the person not to enter the area. It also captures the person's image and temperature and saves the information in a database folder with a time stamp.

If a person's temperature is found to be less than 37 (or say, up to 37) degrees Celsius, the system allows the person to enter after proper sanitisation. The person's image is also saved in a separate database folder, which includes name and temperature along with time stamp.

Fig. 6 shows the code for the image (taken from the captured video) along with the name and temperature. The outline of the rectangular frame showing the name and image will change its colour depending on the detected temperature (red = high temperature, green = low temperature).

Connection

After coding, connect the MLX sensor as shown in Table 2. Connection diagram of MLX90614 with RPI is shown in Fig. 8. Connect the camera, HDMI display, and sensor as shown in the author's prototype (Fig. 7).

Testing

After setup, power the Raspberry Pi and mount the device on a wall or the entry door. Rerun the code and wait for a few seconds. It will show the temperature and face of the person standing at the entrance in front of the camera. If the person is detected to have a high temperature/fever, the device will warn the person not to enter.